

A Neutrino Corridor for a 10 TeV Muon Collider at Fermilab

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The decay neutrinos of a multi-TeV muon collider form pencil beams of unprecedented intensity and energy, and their interaction with the surrounding landscape — where they graze the Earth’s surface — is a recognized siting constraint. We invert the constraint into an asset. We show that an 11 km collider ring fits the Fermilab campus with its interaction-point straight on the true-north meridian -88.222972° , tilted upward by 15.4 mrad (one LEP of ring-plane inclination). The north-going neutrino beam then leaves the ground *inside the laboratory fence*, delivering $\sim 3 \times 10^{10}$ TeV-scale neutrino interactions per tonne-year to an on-site experiment on the present IMCC-class optics — whose beam divergence, not the decay kinematics, sets the spot size [4] — and $\sim 2.6 \times 10^{12}$ if a dedicated low-divergence drift is built, before climbing a utility right-of-way into federally navigable airspace; the south-going beam dives beneath the region’s aquifers to a perigee of 789 m and resurfaces 198 km away on University of Illinois agricultural land, where its exit strip doubles as a second neutrino experiment of $\sim 10^6$ – 10^8 interactions per tonne-year. The second interaction point’s south beam exits onto the same university’s airport. Every point at which a beam passes within 500 ft of the ground surface lies on federal, utility-easement, or university land, and every potable-aquifer crossing beyond the laboratory fence is bounded at a factor $\gtrsim 5 \times 10^3$ ($\gtrsim 45$ even with the dedicated drift) below drinking-water limits. The alignment costs $\sim 11\%$ of one accelerator ring’s circumference and one LEP-scale plane tilt, and converts the muon collider’s defining radiological liability into two shared-science facilities.

Introduction.—A muon collider at $\sqrt{s} = 10$ TeV is the most compact route to the energy frontier now under serious study [1, 2]. Its defining environmental feature is also its most distinctive experimental opportunity: every muon decays, and the decays in each long straight section emit neutrinos along the straight’s axis — a cone of half-angle $1/\gamma \simeq 21 \mu\text{rad}$ (at $E_\mu = 5$ TeV) for a FODO straight, broadened to an effective $\sigma_\theta \simeq 0.15$ mrad for the interaction-point straight by the final-focus optics [4]. Where that plume intersects the Earth’s surface tens of kilometres away, the showers from neutrino interactions in the final metres of soil can produce annual doses approaching regulatory limits — the “neutrino radiation” problem first quantified by King [3] and adopted as a driving constraint by the International Muon Collider Collaboration, whose reference siting aims its insertion plumes at the Mediterranean and a fenced mountain-side [2]. The same beam, sampled *before* it exits, is the most intense high-energy neutrino beam ever conceived, with fully known flavor composition and parent kinematics [5].

This Letter reports a complete siting geometry for the Fermilab campus in which both properties are engineered simultaneously and deliberately: the beam’s two ground intersections are *chosen*, both fall on institutional land, both host experiments, and the intervening 200 km of private land is crossed only at altitudes or depths at which no surface owner’s use of their property is touched.

The geodetic coincidence.—Three surveyed facts make the geometry work. (i) A commercial transmission right-of-way (ComEd *Aurora–Wayne*) runs almost exactly due north–south at longitude $\simeq -88.223^\circ$ from the Fermilab boundary northward. (ii) The northeast corner of

the laboratory’s site polygon lies on that meridian to within one metre of longitude. (iii) Along the meridian -88.222972° the site offers 5.48 km of north–south extent — room for an 11 km racetrack’s 700 m straights and 1.5 km-radius arcs, and for the acceleration chain’s straights on the same line. Because a due-north geodesic holds constant longitude, aligning every long straight with the corridor reduces to placing it on this meridian at azimuth 0° .

Machine geometry.—We fit an 11.0 km racetrack collider ($R_{\text{arc}} = 1528$ m, 2×700 m insertions) against the real site boundary with a 150 m setback, its east straight on the corridor meridian and its interaction point (IP) at 41.84428° N, placed such that both ends of the straight and both beam emergences remain well inside the boundary. The ring plane is tilted $\theta_0 = 15.40$ mrad (0.88°) upward toward north — 10% beyond LEP’s built precedent of 14 mrad — with the straightaway at 191 m above sea level, 34 m below local grade, at the glacial-drift/Silurian-dolomite contact. The rapid-cycling-synchrotron chain (6.28 and 14.72 km racetracks at 60–80 m depth) shares the meridian with mrad-scale plane pitches so that all straight plumes converge on one on-site detector volume; the alignment’s only machine cost is a 10.8% circumference shortfall in the largest RCS relative to an unconstrained boundary-hugging fit.

In a spherical-geoid frame a straight beam launched at height h_0 with elevation angle θ_0 has height $h(s) = h_0 + \theta_0 s + s^2/2R_\oplus$ above the curving surface. Northward, with real (SRTM) terrain, the beam surfaces at 41.8640° N — 2.2 km from the IP and 655 m *inside* the laboratory fence — crosses the boundary 8 m above grade, climbs the right-of-way, passes the nearest resi-

dential area 116 m (380 ft) above grade — 350 ft above a 10 m roofline — and enters federally navigable airspace (500 ft above ground) 9.3 km beyond the fence. Southward the same straight aims 15.40 mrad below the horizon: the beam passes beneath the Aurora conurbation at 74–152 m depth, reaches a perigee of 789 m in the brackish upper Mt. Simon sandstone — the horizon northern Illinois uses for natural-gas storage, sealed beneath the Eau Claire aquitard and below every potable unit — and re-emerges $2\theta_0$ of central angle later: 198 km south, at 40.067° N, on the University of Illinois Urbana-Champaign (UIUC) *South Farms*, approaching under the city of Urbana at 66–119 m depth. A ± 0.1 mrad trim moves this exit ± 1.3 km along the meridian: targeting a chosen parcel is a survey problem, not a physics one. The diametrically opposite second IP, on the ring’s west straight (meridian -88.2598°), inherits the same plane tilt; its south beam exits at $\simeq 40.045^\circ$ N on Willard Airport — owned and operated by the same university — after a $+0.13$ mrad trim well inside the ± 1 mrad mover budget already carried by the IMCC design [2].

The civic envelope.—The siting rule that organizes the study is geometric: wherever either beam is within 500 ft (152 m) of the ground surface, vertically, the surface land must belong to an institutional partner — the Department of Energy, the utility easement, or the university. Above 500 ft over uncongested land an aircraft may legally fly (14 CFR 91.119); since *United States v. Causby* (1946) a landowner’s airspace claim reaches only the “immediate reaches” they can use, so a silent, invisible, depositing-nothing neutrino pencil hundreds of feet up is comfortably in the public highway. Below 500 ft of rock the beam is deeper than every private well (the shallow Silurian aquifer’s wells run 30–90 m), and — unlike the deep tunnels bored under Chicago through thousands of recorded subsurface easements — nothing is built: neutrinos simply pass. Auditing the baseline against this rule leaves four honest exceptions, all deep or high crossings, none touching the surface: 1.9 km of overflight at 384–500 ft over low-density countryside; 5.6 km at 74–152 m under Aurora’s east edge; and the farmland-plus-Urbana approach at 66–152 m. Standard aviation or subsurface easements close each, though none plausibly requires it.

Fluxes and rates.—With IMCC-class beam parameters (1.8×10^{12} muons per bunch per sign at 5 Hz, 1.2×10^7 s operational year, 90% transmission, and 85% of stored muons decaying within each 0.2 s store), 5.3×10^{18} decays per year are aimed along each direction of the IP straight (700 m of an 11 km ring). The pencil-limit on-axis flux is $\Phi = N\gamma^2/\pi L^2$, but on the present optics the plume is divergence-smeared: MINT [4], decaying muons along the IMCC v0.6+v0.9 interaction-region lattice, finds $\sigma_\theta \simeq 0.15$ mrad $\simeq 7/\gamma$, diluting the on-axis density by $2\gamma^2\sigma_\theta^2 \simeq 10^2$ and washing out the energy-angle correlation, so any point in the core sees the angle-

integrated spectrum, $\langle E_\nu \rangle \simeq 0.33 E_\mu$. An on-site detector 1.3 km from the IP then intercepts $\sim 3 \times 10^{10}$ charged- and neutral-current interactions per tonne-year on axis (CSMS cross-sections, both species per decay) — six orders of magnitude beyond present TeV-neutrino samples, in a beam whose flavor content ($\nu_\mu \bar{\nu}_e$ or $\bar{\nu}_\mu \nu_e$, selectable by circulation direction) is exactly known — and $\sim 2.6 \times 10^{12}$ if a fraction $f \simeq 0.5$ of the straight is rebuilt as a dedicated dispersion-free drift (a waisted $\beta^* \sim \ell$ optic with $\sigma_\theta = \sqrt{\epsilon/\ell} \simeq 1 \mu\text{rad}$), the one machine-design ask this concept adds. At the South Farms exit, 198 km out, the surviving flux delivers $\sim 1.3 \times 10^6$ ($f = 0$) to $\sim 10^8$ ($f \simeq 0.5$) interactions per tonne-year: the exit strip that must be controlled anyway becomes a far detector site with L/E two orders of magnitude beyond any accelerator baseline, on land owned by a research university. The aligned RCS chain adds a time-gated 63 GeV–5 TeV energy chirp carrying $\sim 40\%$ of the collider’s rate at the deep-hall option, since its FODO straights emit true $1/\gamma$ pencils [6].

Beam size.—Muon decay is isotropic in the muon rest frame, so a single decaying muon emits the beamed profile $dN/d\Omega = N\gamma^2/[\pi(1 + \gamma^2\theta^2)^2]$ about its own direction. Along the IP straight those directions are not the axis: a field-free drift conserves angles, the final-focus drifts are ~ 150 m long at $\sigma_\theta = \sqrt{\epsilon_N/\gamma\beta^*} = 0.59$ mrad ($\epsilon_N = 25 \mu\text{m}$, $\beta^* = 1.5$ mm), and the matching sections contribute dispersively, so the full-lattice simulation of Ref. [4] finds an effective plume divergence of 0.1–0.2 mrad $\gg 1/\gamma$ with the energy-angle correlation (the “prism”) washed out. This study adopts $\sigma_\theta = 0.15$ mrad: the 50% radius is then 24 cm at the on-site detector, 52 cm at the fence, and 36 m at the southern exit (the kinematic 2.7 cm / 4.2 m survive only for the dedicated-drift pencil fraction f). A one-tonne tungsten cylinder no longer swallows the beam — at $f = 0$ it intercepts 8% — so the on-site detector is better modelled on the low-density tracking-target benchmarks of Ref. [4], sized to the core with a front monitor for the ~ 2 rock muons per bunch crossing that reference computes. At the exits the cross-section is stretched along the ground by $1/\sin(\text{graze})$: the northern emergence paints a 0.8 m $\times$ 50 m streak inside the laboratory fence, while the southern one is a 72 m $\times$ 4.6 km strip, which is the actual geometry of the land-use question — wider than the pencil model gave, with its peak dose diluted by the same factor of $\sim 10^2$.

Radiological audit.—Dose: everywhere along the corridor except the two chosen exits the beam is deep underground or high overhead, and the exit strips themselves are the entire land-use footprint; at the southern exit the $1/L^2$ dilution over 198 km and the measured plume divergence bring even King’s deliberately conservative equilibrium model [3] to ~ 0.8 mSv/yr with no further mitigation (~ 0.2 mSv/yr with a ± 0.5 mrad vertical segmentation) on fenced farmland; the dedicated-drift pencil would con-

concentrate ~ 40 mSv/yr into a 10 m core there, making detector rate and exit dose a single design knob (the framework, corrections, and the deep-reference alternative are in the accompanying study [6]). Groundwater: neutrinos activate nothing directly; the only pathway is the hadronic showers of the $\sim 2 \times 10^{15}$ interactions per year distributed along the full 198 km chord (3.5×10^{-4} interaction lengths of rock), yielding $\sim 6 \times 10^{16}$ tritium atoms per year ($\simeq 0.05$ Ci/yr at saturation) with a $1/L^2$ weighting toward the IP. In the maximally conservative limit — groundwater stagnant inside the decimetres-wide beam core at each aquifer’s closest approach, for many tritium lifetimes, with zero exchange — the EPA drinking-water limit (740 Bq/L) is exceeded only within ~ 0.2 km of the IP (~ 1.4 km in the dedicated-drift scenario), entirely inside the laboratory boundary. Traced through a dipping stratigraphic model of the Illinois Basin’s flank (unit tops anchored to NuMI construction geology, ISGS mapping, and the region’s gas-storage fields), the municipal sandstone aquifers are crossed ~ 14 km and ~ 28 km out and ceiling at $\sim 5 \times 10^3$ and $\sim 2 \times 10^4$ below the limit ($45\times$ and $190\times$ with the dedicated drift); the chord bottoms in the brackish Mt. Simon; and the drift sands near Champaign–Urbana ceiling at $\sim 10^6\times$ below. Groundwater advection — metres per day in the karstic on-site dolomite — lowers all of these by further orders of magnitude.

Conclusion.—The neutrino radiation problem of a 10 TeV muon collider at Fermilab can be solved by construction rather than mitigation: one LEP of ring-plane tilt and a meridian alignment place both surface intersections of the collision straight — and both of the second in-

teraction point’s — on federal or university land, thread the 200 km between them outside the envelope of private use, bound aquifer activation orders of magnitude below drinking-water standards, and produce two neutrino experiments of unprecedented reach as a by-product. The geometry uses only surveyed data and analytic conservative models; a real lattice, FLUKA/MARS transport, and hydrogeological modelling are the obvious next steps, and all inputs, code, and an interactive reproduction of every number in this Letter are public [6].

Geometry, terrain, and boundary data: OpenStreetMap contributors (ODbL) and NASA SRTM. This concept study used no beam time and harmed no groundwater.

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